

Digital Parking Permit System - SER216 Final Project

GitHub Repository Link: https://github.com/elzodxon/SER216_Final_Project_Elzodxon

Task 1: Requirements

Functional Requirements:

1. The system shall allow students and staff to register vehicles using plate number, owner ID, and vehicle type.
2. The system shall allow users to request parking permits for available parking zones.
3. The system shall display available parking zones before permit approval.
4. The system shall allow security staff to verify permits and update violation records.

Non-Functional Requirements:

1. The system shall verify parking permits within 3 seconds during security checks.
2. The system shall protect user and vehicle data using secure authentication and access control.

Task 2: Workflow Diagram

Vehicle Registration -> Permit Request -> Zone Check -> Permit Approval -> Parking Verification

The updated workflow diagram is included on the next page and stored at diagrams/workflow_diagram.png.

Task 3: Test Case Table

TC01 System Test: Complete permit process with valid student, vehicle, and zone. Expected: approved permit.

TC02 System Test: Security verifies an approved valid permit. Expected: permit is valid.

TC03 Acceptance Test: Student requests permit for available zone. Expected: approved by university rules.

TC04 Acceptance Test: User checks parking zones. Expected: correct available zones shown.

TC05 Regression Test: Valid non-expired permit. Expected: not shown as expired.

TC06 Regression Test: Security adds violation. Expected: saved without affecting permit status.

Task 4: Defect Analysis

Defect: A valid parking permit is shown as expired during security verification.

Defect category: Functional defect.

Severity: High.

Priority: High.

Possible root cause: Incorrect date comparison, timezone mismatch, database expiry mismatch, or cached permit status.

Regression test: Verify that a valid permit with a future expiry date is displayed as valid during security verification.

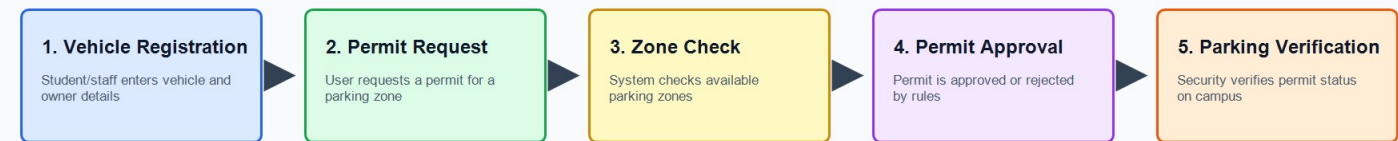
Task 5: GitHub Evidence

The following pages contain screenshots from the GitHub repository showing the required evidence.

Workflow Diagram

Digital Parking Permit System

Workflow Diagram



Main flow: Vehicle Registration -> Permit Request -> Zone Check -> Permit Approval -> Parking Verification

Actors: students, staff, system administrator, and security staff

Repository Homepage



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final_report.pdf

README

SER216 Final Project - Digital Parking Permit System

Project Description

This project describes a Digital Parking Permit System for a university. Students and staff can register vehicles, request parking permits, check available parking zones, and report parking issues. Security staff can verify permits and update violation records.

Tools Used

- diagrams.net style workflow diagram
- Google Sheets compatible test case table
- PDF final report
- GitHub for version control, branching, pull request evidence, and final submission

About

SER216 Final Project - Digital Parking Permit System

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Activity

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Report repository

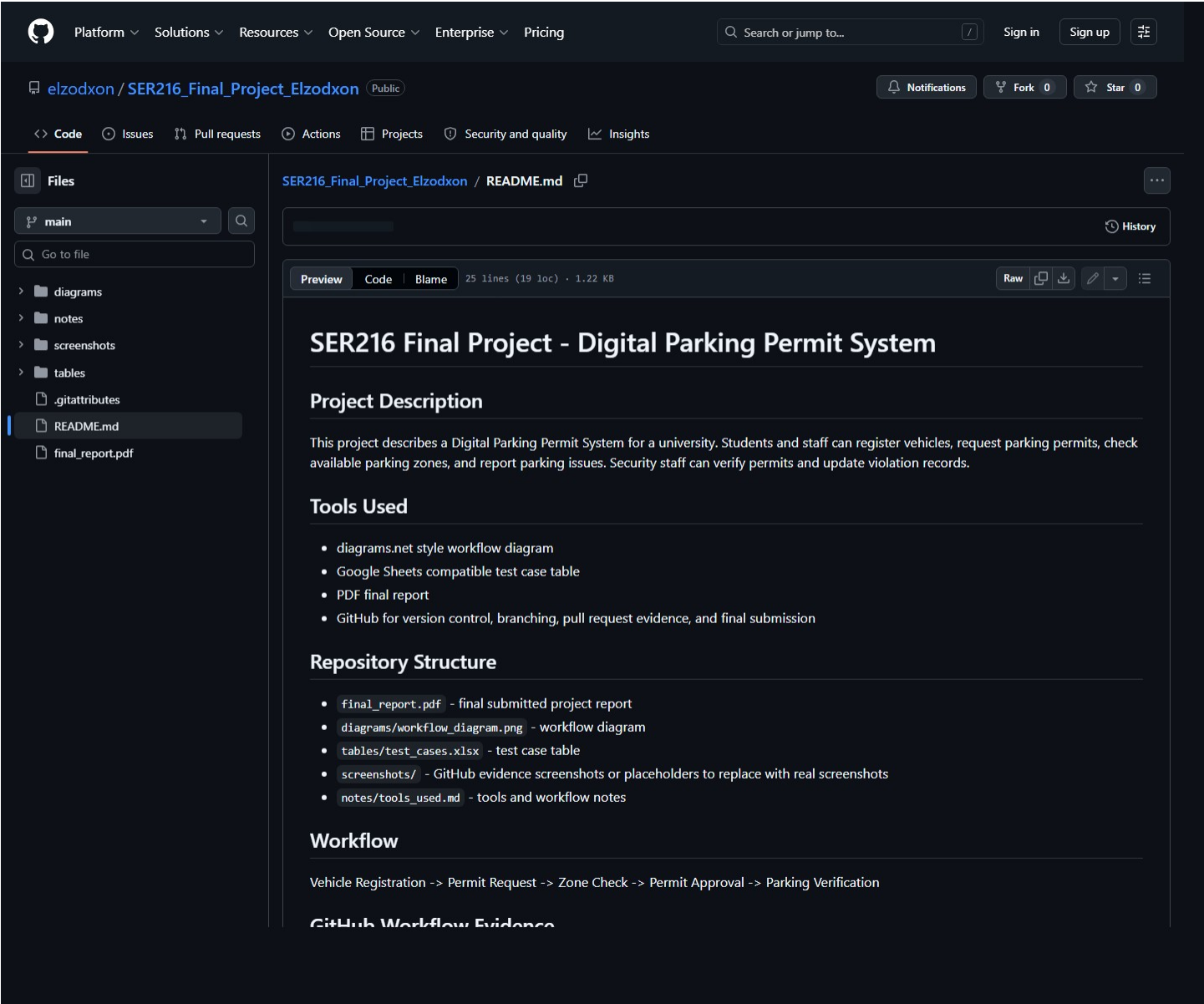
Releases

No releases published

Packages

Contributors

README.md File



Commit History - 3+ Commits



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Commits

main

All usersAll time

Commits on May 4, 2026

Update repository link in final materials

Elzodxon committed 3 minutes ago

99d8a68

Merge pull request for SER216 final project

elzodxon authored 5 minutes ago

557a984

Add final report and GitHub evidence screenshots

Elzodxon committed 9 minutes ago

e1aa58f

Add workflow diagram and test case table

Elzodxon committed 16 minutes ago

af8fa3e

Add project README and tools notes

Elzodxon committed 16 minutes ago

589c077

Feature Branch

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Branch	Updated	Check status	Behind Ahead	Pull request
main	3 minutes ago			

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feature/test-plan	8 minutes ago			

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Add final SER216 project materials #1

Merged

elzodxon merged 2 commits into main from feature/test-plan 5 minutes ago

Conversation 0

Commits 2

Checks 0

Files changed 10

+3

elzodxon commented 5 minutes ago

Owner

...

Adds workflow diagram, test case table, final report, and GitHub evidence screenshots.

Elzodxon added 2 commits 16 minutes ago

Add workflow diagram and test case table

af8fa3e

Add final report and GitHub evidence screenshots

e1aa58f

elzodxon merged commit 557a984 into main 5 minutes ago

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Successfully merging this pull request may close these issues.

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1 participant

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99d8a68 · 3 minutes ago

History

4.19 KB